PAPER_411 – Ug1: Di-Pseudo-Monopole (DPM) Internal Dipole — Solar Calibration and Defect Mechanism

***Session:** 110 (grokshare755feea7.txt analysis)*

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Source: grok*share*755feea7.txt — "Star Magic" Chapter 1, Section Ug1, Solar Refinement **Session:** 110 (grok*share*755feea7.txt analysis) **CP4 Class:** `Ug1DPMDiPseudoMonopoleSolarCalibrationCalculator` (#61)

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1. Overview

PAPER_411 establishes **Ug1 — the Di-Pseudo-Monopole (DPM)** as the foundational internal driving force of any star. Ug1 is the **first and primary** gravity range in the UQFF hierarchy, capturing the stellar dipole moment arising from the interaction of Universal Aether derivatives (UA') with trapped SCm.

Key contributions of Ug1:

- Drives surface irregularities through internal defects (δ_{def})
- Is the **origin force** from which Ug2, Ug3, Ug4, and Ug4i all cascade
- Is modulated by π cycles and non-linear time decay
- Has been calibrated using specific solar data: $\mu_s \approx 3.38 \times 10^{20} \, \text{T·m}^3$ (base)

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2. Ug1 Equation

$$Ug_1 = k_1 \cdot \mu_s(t, \rho_{\text{vac},[\text{SCm}]}) \cdot \nabla\!\left(\frac{M_s}{r}\right) e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + \delta_{\text{def}})$$

where:

Symbol	Value (Sun)	Description
k_1	1.5	Coupling constant for Ug1 (refined)
μ_s	$(10^3 + 0.4\sin(\omega_c t)) \cdot 3.38 \times 10^{20}$ T·m^3	Stellar DPM moment with SCm
$\nabla(M_s/r)$	≈ 274 m/s^2	Gradient of gravitational potential at $r = R_s$
α	0.001 day^{-1}	Non-linear time decay rate
$\cos(\pi t_n)$	oscillatory	π cycle modulation with negative time
δ_{def}	$0.01 \cdot \sin(0.001 \, t)$	Defect factor — drives surface irregularities

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3. Di-Pseudo-Monopole Physics

3.1 DPM Definition

The Di-Pseudo-Monopole is identified as:

$$\text{DPM} \equiv \frac{[UA']}{[\text{SCm}]}$$

where $[UA']$ is the first-order derivative of the Universal Aether, meaning the **gradient flux** of Aether as it interacts with the internal SCm configuration. The monopole character arises because adjacent field lines cannot escape (monopole-like), but the system is an internal dipole (hence "di-pseudo").

3.2 Stellar DPM Moment (μ_s)

For a star, the DPM moment is:

$$\mu_s(t, \text{SCm}) = \left[B_s(t) + B_{\text{SCm}}\right] \cdot R_s^3$$

where:

- $B_s(t) = B_s^{(0)} + 0.4 \cdot \sin(\omega_c t)$ — time-varying surface field
- $B_{\text{SCm}} \approx 10^3$ T — SCm-driven superconductive contribution (undetectable via $Q_s = 0$)

Solar Values:

$$B_s^{(0)} = 10^{-4} \, \text{T}, \quad R_s = 6.96 \times 10^8 \, \text{m}$$

Base DPM moment (no SCm):

$$\mu_{s,\text{base}} = 10^{-4} \cdot (6.96 \times 10^8)^3 \approx 3.38 \times 10^{20} \, \text{T·m}^3$$

Full DPM moment (with SCm):

$$\mu_{s,\text{full}} \approx (10^3 + 0.4\sin(\omega_c t)) \cdot 3.38 \times 10^{20} \approx 3.38 \times 10^{23} + 1.35 \times 10^{20} \sin(\omega_c t) \, \text{T·m}^3$$

The **SCm contribution dominates** by 7 orders of magnitude over the bare magnetic field.

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4. Gravitational Gradient Term

$$\nabla\!\left(\frac{M_s}{r}\right) \approx \frac{G M_s}{R_s^2} = \frac{6.674 \times 10^{-11} \cdot 1.989 \times 10^{30}}{(6.96 \times 10^8)^2} \approx 274 \, \text{m/s}^2$$

This is the **surface gravity** of the Sun, confirming dimensional alignment of the DPM gradient term.

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5. Numerical Calibration at t = 0

With $k_1 = 1.5$, $\delta_{\text{def}} = 0$, $\cos(\pi \cdot 0) = 1$:

$$Ug_1(t=0) \approx 1.5 \cdot 3.38 \times 10^{23} \cdot 274 \cdot 1 = 1.39 \times 10^{26} \, \text{[normalized units]}$$

With solar cycle oscillation:

```
Ug_1(t) \approx (1.39 \times 10^{26} + 5.55 \times 10^{23} \cdot \sin(\omega_c t)) \cdot e^{-0.001t} \cdot \cos(\pi t)
```

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6. Surface Irregularities via δ_def

The defect factor models surface phenomena (sunspots, flares, magnetic anomalies):

```
\delta_{\text{def}}(t) = 0.01 \cdot \sin(0.001 \, t)
```

This modifies Ug1 by ±1% with a period of roughly 6,280 seconds (~1.7 hours), representing **rapid surface defect cycles** driven by the internal SCm structure.

The Sun's observable surface activity (sunspots, differential rotation patterns) is a **direct readout** of these Ug1 internal defects — the surface magnetism is **unique to the surface**, not the interior.

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7. Cascade to Ug2, Ug3, Ug4

Ug1 is the generative force for all higher Ug terms:

Cascade Effect	Mechanism
Ug1 → Ug2	DPM field bubble expands outward, forming the heliosphere via charge-reactive Aether trapping
Ug1 → Ug3	Equatorial CCW vs coronal CW spin differential (arising from DPM asymmetry) creates the magnetic strings disk
Ug1 → Ug4	DPM field propagates to the star–black hole interaction scale via vacuum energy modulation
Ug1 → Ug4i	Sub-range DPM effects extend to galactic vacuum fluctuation coupling

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8. Code Implementation

```
double compute_mu_s(double t, double Bs, double omega_c, double Rs, double SCm_contrib = 1e3) {
double Bs_t = Bs + 0.4 * std::sin(omega_c * t) + SCm_contrib;
return Bs_t * std::pow(Rs, 3);
}
double compute_grad_Ms_r(double Ms, double Rs) {
if (Rs <= 0.0) throw std::runtime_error("Invalid Rs value");
return G * Ms / (Rs * Rs);
}
double compute_Ug1(const CelestialBody& body, double r, double t, double tn,
double alpha, double delta_def, double k1) {
if (r <= 0.0) throw std::runtime_error("Invalid r value");
double mu_s = compute_mu_s(t, body.Bs_avg, body.omega_c, body.Rs);
double grad_Ms_r = compute_grad_Ms_r(body.Ms, body.Rs);
double defect = 1.0 + delta_def * std::sin(0.001 * t);
return k1 * mu_s * grad_Ms_r * std::exp(-alpha * t) * std::cos(PI * tn) * defect;
}
```

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9. Unit Test

```
def test_Ug1_solar_calibration():
"""Solar Ug1 at t=0, no defect"""
import math
G = 6.674e-11; Ms = 1.989e30; Rs = 6.96e8
Bs = 1e-4; SCm_contrib = 1e3; k1 = 1.5; alpha = 0.001; t = 0; tn = 0
mu_s = (Bs + SCm_contrib) * Rs**3 # ≈ 3.38e23
grad_Ms_r = G * Ms / Rs**2 # ≈ 274
defect = 1.0
expected = k1 * mu_s * grad_Ms_r * math.exp(-alpha * t) * math.cos(math.pi * tn) * defect
# ≈ 1.39e26
assert expected > 1e25, f"Ug1 solar calibration failed: {expected}"
```
